

MCCOOK, NE, USA

WMO: 725625

Lat: 40.206N

Lon: 100.591W

Elev: 771

StdP: 92.40

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.8	-21.8	0.6	-16.5	-19.0	0.8	-13.3	14.9	5.5	12.9	5.4	3.7	250	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	37.8	21.4	36.1	21.3	34.0	21.1	24.1	32.0	23.2	31.6	22.4	30.9	6.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.3	28.3	21.0	17.2	27.1	20.0	16.1	26.0	76.5	31.9	73.0	31.6	69.7	30.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.5	10.0		-22.4	40.9	2.7	1.9	-24.3	42.3	-25.9	43.4	-27.5	44.5	-29.4	45.9	
			DB	-22.7	25.4	2.6	0.7	-24.6	25.8	-26.2	26.2	-27.6	26.6	-29.6	27.1	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-1.4	-0.2	5.7	10.6	16.4	22.6	25.8	24.1	19.7	11.6	4.6	-1.4	(6)
	DBStd	10.96	6.09	6.29	5.87	5.29	4.82	3.94	3.28	3.14	4.71	5.32	5.42	5.86	(7)
	HDD10.0	1428	352	288	157	54	7	0	0	0	1	44	173	352	(8)
	HDD18.3	3141	611	519	392	235	96	8	1	2	39	217	412	610	(9)
	CDD10.0	2007	1	2	23	73	204	378	491	438	293	93	11	0	(10)
	CDD18.3	679	0	0	0	4	35	137	233	182	81	7	0	0	(11)
	CDH23.3	8402	0	2	38	126	542	1668	2825	1979	1048	162	10	0	(12)
	CDH26.7	4190	0	0	7	34	213	821	1567	1006	492	49	1	0	(13)
(14) Wind	WSAvg	4.5	4.3	4.6	5.0	5.6	5.0	4.6	4.0	3.7	4.2	4.5	4.3	4.3	(14)
(15) Precipitation	PrecAvg	521	12	15	34	49	77	82	84	66	40	31	22	12	(15)
	PrecMax	750	38	67	107	147	176	198	201	156	181	124	95	50	(16)
	PrecMin	246	0	0	0	3	0	11	18	3	2	1	0	0	(17)
	PrecStd	103	11	15	27	35	45	45	45	38	38	27	21	11	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	21.9	27.4	30.7	34.9	38.9	40.8	38.9	37.3	32.0	24.9	17.4	(19)
		MCWB	7.6	9.8	13.5	14.4	17.9	21.1	21.3	20.6	19.8	17.1	12.2	7.9	(20)
	2%	DB	14.4	17.0	23.6	27.0	31.7	36.5	38.1	36.3	34.2	27.6	21.1	13.2	(21)
		MCWB	6.2	7.7	11.3	14.3	18.0	21.0	21.9	21.1	19.8	15.6	10.8	5.8	(22)
	5%	DB	11.1	12.9	20.1	23.9	28.8	33.9	36.4	34.1	32.3	24.4	17.5	10.6	(23)
		MCWB	4.6	5.9	10.2	13.3	17.3	20.8	22.2	21.6	19.3	14.2	9.2	4.6	(24)
	10%	DB	7.8	9.8	17.0	20.9	26.3	31.4	34.1	32.4	29.2	21.5	14.0	7.5	(25)
		MCWB	2.8	4.5	8.8	12.2	16.4	20.2	22.0	21.4	18.2	13.1	7.2	2.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.3	15.5	16.9	21.2	24.2	25.2	24.9	22.8	19.1	13.6	9.0	(27)
		MCDB	17.9	18.7	21.8	25.9	29.4	31.6	33.8	32.7	31.3	26.5	20.6	15.3	(28)
	2%	WB	6.4	8.1	12.9	15.4	19.6	22.8	24.4	23.7	21.3	17.1	11.5	6.4	(29)
		MCDB	14.0	15.7	20.2	23.7	28.0	32.1	32.6	31.5	30.2	23.9	18.4	13.1	(30)
	5%	WB	4.8	6.4	10.8	14.0	18.5	21.9	23.6	22.8	20.4	15.5	9.7	4.7	(31)
		MCDB	10.8	12.4	19.1	21.9	26.4	30.8	31.9	30.7	28.8	22.4	16.9	10.2	(32)
	10%	WB	3.0	4.6	9.0	12.7	17.3	21.0	22.8	22.0	19.5	13.9	7.8	2.9	(33)
		MCDB	7.3	9.7	16.4	20.1	24.3	29.7	31.4	29.6	27.3	20.2	14.1	7.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	13.2	15.5	14.7	14.4	14.6	14.5	14.2	15.6	15.1	15.0	13.1	(35)
		MCDBR	19.0	19.4	22.2	20.5	18.5	18.1	17.6	17.2	18.8	20.5	20.6	18.1	(36)
	5% WB	MCWBR	11.8	11.7	11.6	10.0	7.8	6.4	5.8	6.1	6.9	9.7	11.5	11.5	(37)
		MCWBR	18.1	18.8	19.8	17.3	15.7	15.1	14.7	15.9	16.6	19.6	19.6	17.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.263	0.270	0.297	0.328	0.357	0.367	0.390	0.378	0.346	0.305	0.271	0.261	(40)
		taud	2.542	2.531	2.471	2.408	2.401	2.407	2.384	2.405	2.449	2.538	2.554	2.539	(41)
	Ebn at Noon		923	963	962	946	920	907	882	885	896	906	904	897	(42)
		Edh at Noon	71	83	99	113	117	117	118	113	101	82	70	66	(43)
(44) All-Sky Solar Radiation	RadAvg		2.38	3.24	4.44	5.46	6.31	7.19	7.08	6.16	5.01	3.62	2.63	2.07	(44)
	RadStd		0.18	0.25	0.36	0.29	0.41	0.38	0.27	0.33	0.26	0.36	0.17	0.14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page